

JKBOSE 2026

CLASS 11TH

Morphology of Flowering Plants

LECTURE-04

BOTANY

ARSHID BHAT SIR

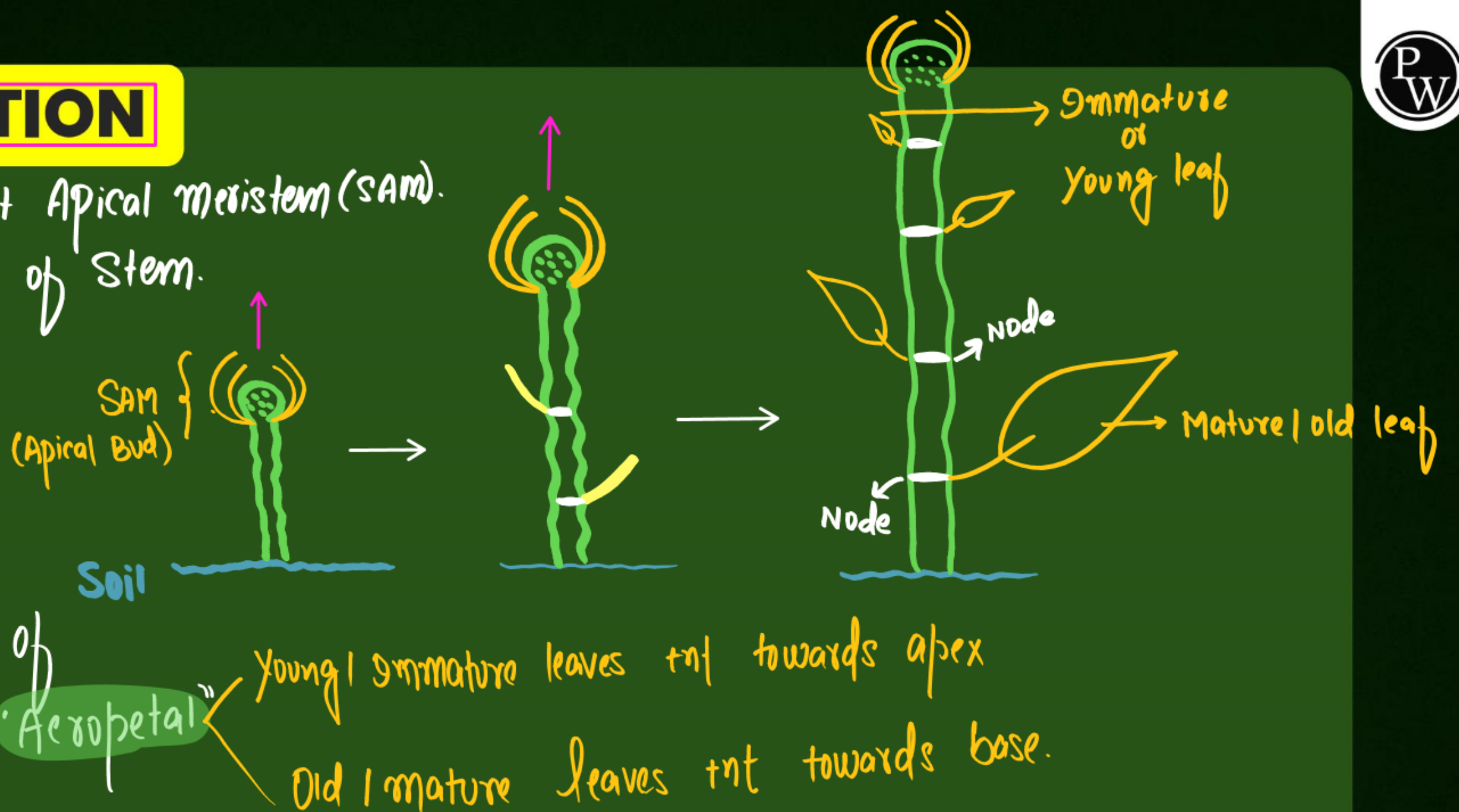


Topics To Covered

- 1** MORPHOLOGY OF LEAF
- 2** ...

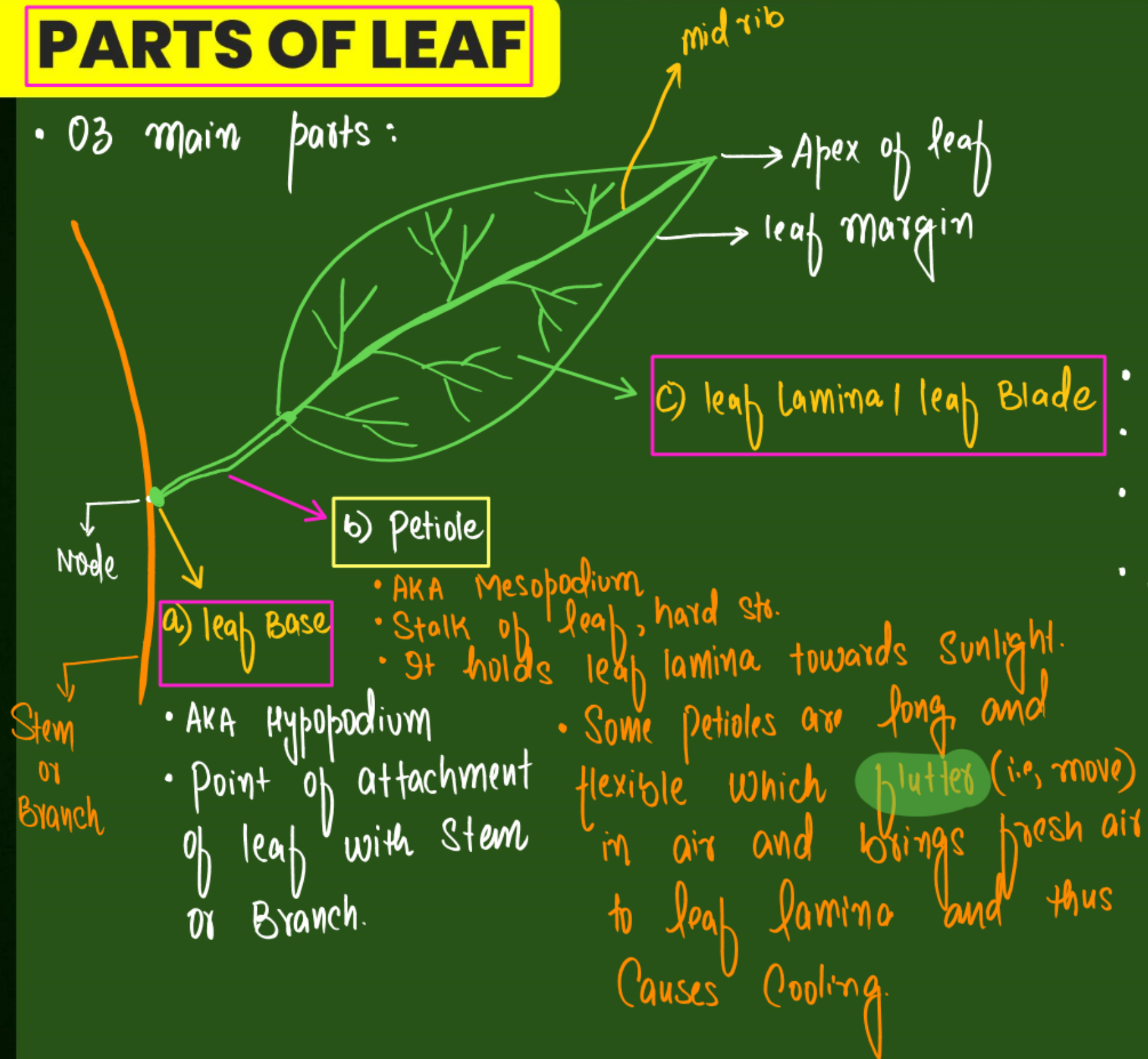
LEAF INTRODUCTION

- It develops from Shoot Apical meristem (SAM).
- It is not at node of stem.
- Leaves are borne laterally on stem.
- Perform photosynthesis.
- Order of arrangement of leaves on stem is "Acropetal"



PARTS OF LEAF

- 03 main parts:



- AKA Epipodium
- Green, flattened structure, performs photosynthesis.
- largest part of leaf.
- It possess **Veins** and veinlets
 - provide mechanical support to leaf lamina.
 - Contains vascular bundle (xylem + phloem) for transport of water, minerals and food.

VENATION



- It is the pattern of arrangement of veins and veinlets inside leaf lamina.
- Two types :

a. Parallel venation :



- Veins run parallel to each other.
- Ex. Mostly in Monocots.

b. Reticulate Venation :

- Veins form network like structure.
- Ex. Mostly in Dicots.



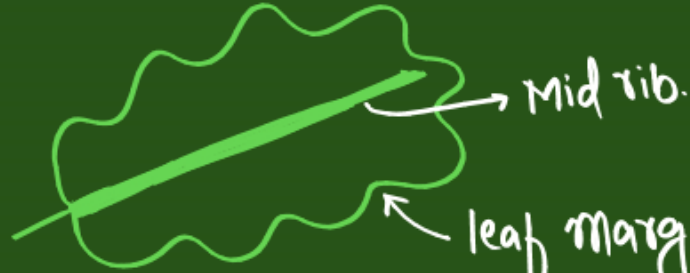
LEAF TYPES

Two main types

① Simple leaf



leaf margin = Entire / Smooth.



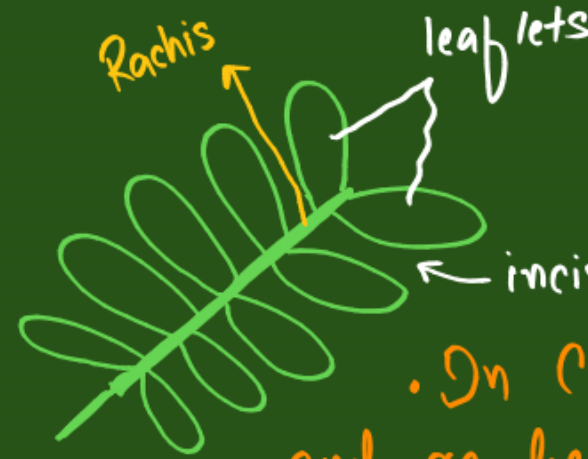
Mid rib.

leaf margin = incision + nt but not upto midrib.

• In Simple leaf, leaf margin is smooth or incised but incisions does not reach upto midrib.

• Ex. Mango, Peepal.

② Compound leaf



• In Compound leaf, incisions are complete and reaches upto midrib and thus leaf lamina gets divided into many leaflets.

• Here midrib is known called as **Rachis**.

• 2 types of compound leaf:

a) Pinnately compound leaf

• Common Rachis possess number of leaflets.



Example: NEEM.

b) Palmately compound leaf:

• NO. of leaflets are present at tip of Petiole

• Rachis - nt

• Ex. Bombax (Silk Cotton).



PHYLLOTAXY

- Pattern of arrangement of leaves on stem/branch.
- 3 types:

① Alternate / Spiral Phyllotaxy

- At each node, single leaf is present alternatively.

• Most Common.

• Ex. China rose
Mustard
Sunflower



② Opposite Phyllotaxy

- At each node two leaves are present opposite to each other.

• Ex. Calotropis
Guava

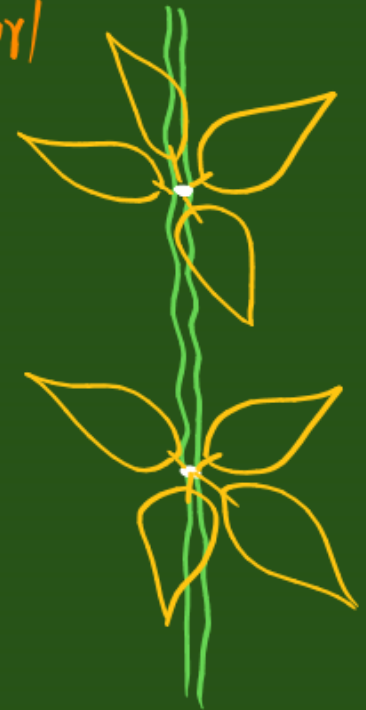


③ Whorled Phyllotaxy

- At each node whorl of leaves are present.

• Rare in nature

• Ex. Alstonia



MODIFICATIONS OF LEAF

- Not in Boards Syllabus
- NEET Syllabus ✓✓

Home work ???

QUESTIONS

Leaves originate from and are arranged in an order.

- A** Root apical meristem, acropetal
- B** Floral meristem, basipetal
- C** Shoot apical meristem, acropetal
- D** Internode, basipetal

HOMework



Modifications of leaf...



Thank
You